

Minimum skills required to get an entry-level Agentic AI job in India

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Research Report: Minimum Skills Required for an Entry-Level Agentic AI Job in India

Executive Summary

The Indian technology landscape is undergoing a major shift from reactive artificial intelligence (traditional LLM chat interfaces) to **Agentic AI**—autonomous systems capable of goal setting, multi-step planning, tool invocation, self-correction, and multi-agent coordination.

With job postings mentioning frameworks like **LangGraph**, **CrewAI**, and **AutoGen** surging over 300% on LinkedIn India, both IT majors (TCS, Infosys, Wipro, Accenture, Deloitte) and fast-growing AI startups are actively hiring entry-level talent. However, landing an entry-level role (such as Junior Agentic AI Engineer or AI Automation Developer) requires more than simple prompt engineering; candidates need a solid foundation in software engineering, API integration, Retrieval-Augmented Generation (RAG), stateful orchestration, and production deployment.

Entry-level compensation in India for freshers and candidates with 0–2 years of experience ranges between **■5 LPA and ■10 LPA** at IT service companies, and **■10 LPA to ■18+ LPA** at AI product startups and product centers for candidates with practical project portfolios.

Key Findings

- Prerequisites are Non-Negotiable:** Recruiters prioritize candidates with strong Python, object-oriented programming (OOP), API consumption, and basic database (SQL/NoSQL) knowledge. Candidates without strong core programming struggle to debug autonomous state loops.
- Shift from Basic LLMs to Stateful Workflows:** Companies require familiarity with **LangGraph** or **CrewAI** over basic LLM API wrappers. Knowledge of graph-based state machines, memory persistence, and human-in-the-loop (HITL) checkpoints is highly valued.
- Tool Calling & RAG as Core Essentials:** Function calling (letting LLMs select and execute external Python tools, SQL queries, or web searches) and Vector DB retrieval (Chroma, FAISS, Pinecone) are baseline requirements for entry-level work.
- Portfolio Trumps Certificates:** Hiring managers in India value GitHub repositories containing deployed, end-to-end agentic applications over theoretical certifications.

Detailed Analysis

Market Demand and Hiring Landscape in India

The demand for Agentic AI talent in India is driven by enterprise adoption. Platforms like Salesforce (Agentforce) and ServiceNow have embedded autonomous agents into production CRM and IT workflows. Consequently, Indian IT services and product firms are building specialized Agentic AI delivery units.

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| Entry-Level Skill Stack |
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| Software Baseline| | Agentic Tech Core|
| - Python (OOP) | | - LLM Function |
| - REST/Async API| | Calling |
| - SQL & Git | | - LangGraph / |
| - Docker & | CrewAI |
| FastAPI | | - RAG / Vector |
+-----+ DBs |
| - Guardrails & |
| Monitoring |
+-----+
```

Minimum Requirements

To be considered job-ready for an entry-level Agentic AI position in India, candidates must possess a blend of foundational software engineering and specialized AI orchestration skills.

1. Foundational Software Engineering

- **Python Proficiency:** Fluent with OOP (classes, inheritance, decorators), data structures (lists, dictionaries, sets), error handling, and asynchronous programming (`asyncio`).
- **API Development & Integration:** Hands-on experience with REST APIs, handling JSON outputs, using `requests` / `httpx`, and building backends with **FastAPI**.
- **Databases & Data Handling:** Proficiency in SQL (PostgreSQL/MySQL) for relational querying, plus familiarity with unstructured data processing (JSON, Markdown, PDF parsing).
- **Version Control & DevOps Basics:** Git/GitHub, environment management (`venv`, `conda`, `uv`), and basic **Docker** containerization.

2. Core Agentic AI Technical Skills

- **LLM APIs & Function/Tool Calling:** Practical experience with OpenAI SDK, Anthropic API, or open-source models via Ollama/HuggingFace. Deep understanding of structured outputs (using Pydantic), JSON mode, and schema definition for function calling.

- **Retrieval-Augmented Generation (RAG):** Document chunking strategies, embeddings, hybrid search (keyword + semantic), and integration with vector databases (**ChromaDB**, **FAISS**, or **Pinecone**).
- **Single & Multi-Agent Frameworks:**
 - **LangChain / LangGraph:** Building directed acyclic graphs (DAGs) and cyclic graphs, managing graph states, edge conditional branching, and memory persistence.
 - **CrewAI / AutoGen:** Setting up role-based multi-agent teams (e.g., Researcher, Editor, Manager), defining delegation rules, and handling hierarchical execution.
 - **Agentic Patterns & Reasoning:** Understanding the **ReAct** (Reasoning + Acting) loop, plan-and-solve execution, human-in-the-loop (HITL) checkpoints, and state back-tracking.
 - **Guardrails & Evaluation:** Validating outputs via **Pydantic** or **Instructor**, managing token costs, avoiding infinite agent loops, and tracking runs using tracing tools like **LangSmith** or **Phoenix**.

Skills Needed (Summary Matrix)

Skill Category	Specific Skills & Tools Required	Importance Level for Freshers
Programming Language	Python 3.10+ (OOP, Async, Type Hinting)	Crucial Baseline
Backend & Deployment	FastAPI, Docker, REST APIs, WebSockets	High
Orchestration Frameworks	LangGraph, CrewAI, AutoGen, LangChain	Core Requirement
Data & Retrieval	RAG, Vector DBs (Chroma, FAISS), SQL, Pydantic	Core Requirement
Model Integration	OpenAI API, Anthropic API, Ollama, Function Calling	Core Requirement
Observability & Safety	LangSmith, Phoenix, Guardrails AI, Infinite loop prevention	Medium to High

Recommended Learning Path

For candidates looking to transition into an entry-level Agentic AI role within 3 to 6 months:

1. **Month 1: Software & API Foundation**
 - Solidify Python (classes, async programming, data parsing).
 - Build REST APIs using FastAPI and containerize them with Docker.
 - Master Git workflows and basic SQL operations.

2. Month 2: LLMs, Function Calling, and RAG

- Work with OpenAI / Anthropic SDKs and local models via Ollama.
- Implement strict Pydantic output validation and native function calling.
- Build custom RAG pipelines using vector stores (ChromaDB / FAISS).

3. Month 3: Single-Agent Systems & LangGraph

- Understand the ReAct loop from scratch in pure Python.
- Learn **LangGraph** primitives: State, Nodes, Edges, StateGraph, and Checkpointers.
- Implement custom tools (e.g., custom Python execution, web scraper, database fetcher).

4. Month 4: Multi-Agent Systems & Production Guardrails

- Build multi-agent workflows using **CrewAI** or **AutoGen** (delegation, agent-to-agent communication).
- Integrate **LangSmith** or **Phoenix** for tracing, logging, cost monitoring, and debugging.
- Deploy agents as API endpoints via FastAPI hosted on cloud platforms (AWS, Azure, Render, or Hugging Face Spaces).

Portfolio / Project Requirements

In the Indian job market, candidates without prior experience must present **2 or 3 non-trivial GitHub projects** to secure interviews. Tutorial-following projects (e.g., basic PDF chatbot) are generally screened out.

High-Impact Project Ideas for Entry-Level Candidates

1. Autonomous Financial/Market Analyst Agent (LangGraph + Tools)

- ***Features*:** Takes a ticker symbol/company name, autonomously queries financial APIs, searches web news, performs sentiment analysis, generates a financial summary report, and posts the report to Slack/Email via API.
- ***Key Demoed Skills*:** Tool integration, multi-step branching, human approval checkpoint before emailing, LangSmith tracing.

2. Multi-Agent Code Review & Debugging Pipeline (CrewAI or AutoGen)

- ***Features*:** A 3-agent crew consisting of a Code Inspector, Security Auditor, and Documentation Generator. Integrates directly with GitHub webhooks to read pull requests and output inline review comments.
- ***Key Demoed Skills*:** Role delegation, multi-agent orchestration, GitHub API integration, structured outputs.

3. Customer Support RAG Agent with Live DB Updates (LangGraph + FastAPI)

- ***Features*:** RAG-powered agent that handles customer queries, fetches live order status from a PostgreSQL database, updates user addresses upon verification, and escalates edge cases to human support queues.
- ***Key Demoed Skills*:** Vector DB retrieval, SQL database tools, human-in-the-loop escalation, stateful graph memory, FastAPI server deployment.

Job Readiness & Market Landscape in India

Target Entry-Level Job Titles

- Junior Agentic AI Engineer
- AI Automation Developer
- LangChain / LangGraph Developer
- Associate Generative AI Developer
- AI Solutions Associate

Expected Salary Ranges in India (2025–2026)

- IT Service Majors (TCS, Infosys, Wipro, Cognizant, Accenture): ■5.0 LPA – ■8.5 LPA
- Mid-Tier IT & Analytics Firms (Fractal, LatentView, Mu Sigma): ■7.0 LPA – ■11.0 LPA
- AI Product Companies & High-Growth Startups (Bangalore, NCR, Hyderabad, Pune): ■10.0 LPA – ■18.0 LPA

Conclusion

To secure an entry-level Agentic AI job in India, candidates do not need a PhD or extensive machine learning research experience. Instead, employers seek **capable software builders** who can reliably engineer autonomous workflows using frameworks like LangGraph and CrewAI, integrate vector databases and custom APIs, apply output validation guardrails, and wrap agent workflows inside production-ready backend APIs. Focusing on these practical competencies alongside a verified portfolio of deployed projects provides the clearest pathway into the field.

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